

CS 3700 — Lab 1: Conceptual Data Modeling with ER Diagrams

Part 1 — Identify Components

Entities

- Patient
- Doctor
- Department
- Appointment

Attributes

- For Patient: PatientID, Name, DateOfBirth(DOB), Phone, Email
- For Doctor: DoctorID, Name, Specialty, OfficeNumber
- For Department: DepartmentID, DepartmentName, Location
- For Appointment: AppointmentID, Date, Time, Reason

Primary Key

- For Patient: PatientID
- For Doctor: DoctorID
- For Department: DepartmentID
- For Appointment: AppointmentID

Relationships

- Patient HAS_AN appointment
- Doctor HAS_AN appointment
- Doctor WORKS_IN department

Part 2 — Determine Constraints

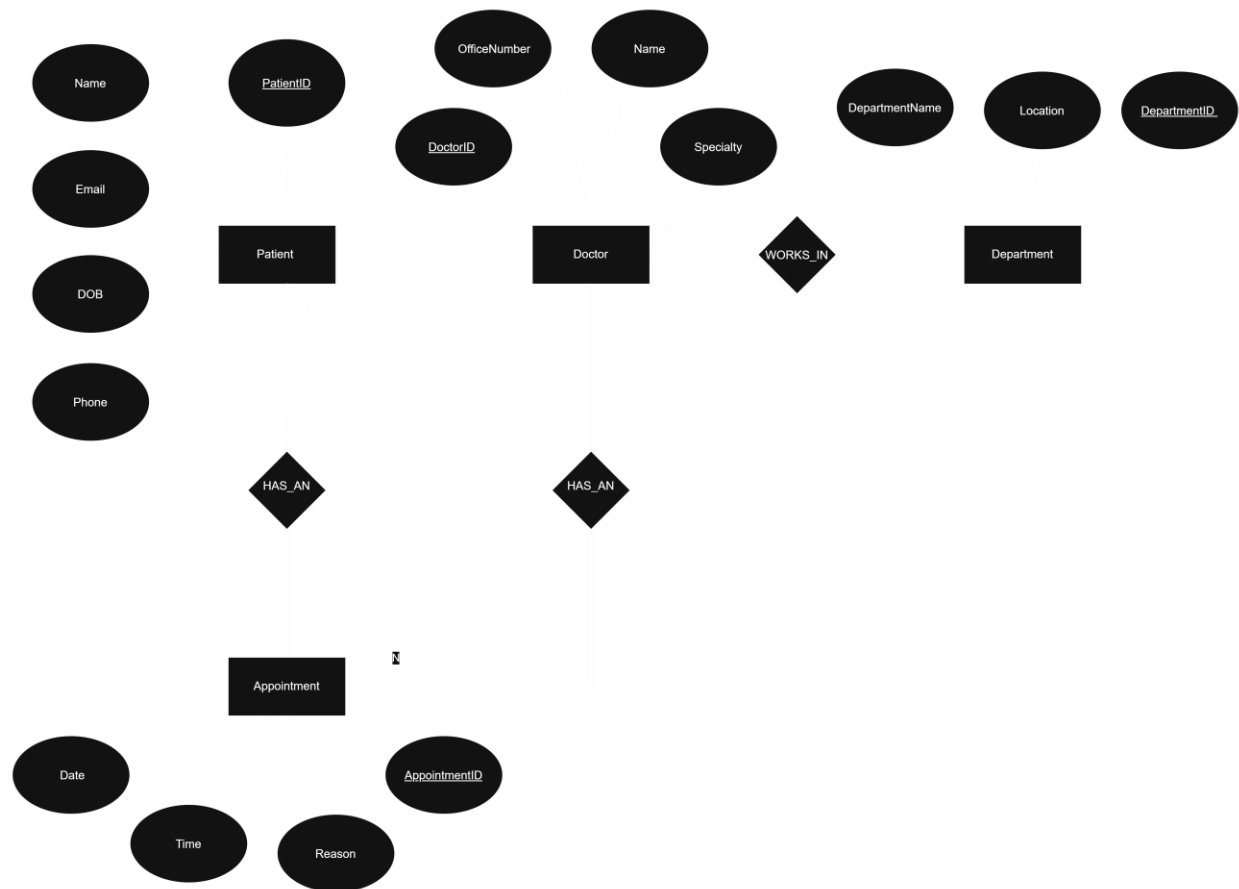
Cardinality

- Patient: Appointment (1: N)
- Doctor: Appointment (1: N)
- Doctor: Department (1: N)

Participation

- Patient(partial): Appointment(total); A patient may or may not have an appointment, but if there is an appointment, there must be a patient
- Doctor(partial): Appointment(total); A doctor may or may not have an appointment, but if there is an appointment, there must be a doctor
- Doctor(total): Department(partial): A doctor must belong to a department to work at a clinic, but department exists always even without a doctor.

Part 3 — Draw the ER Diagram



Draw.io file is also attached with this assignment, the above diagram was copy as image from draw.io

Part 4 — Design Justification

Why is Appointment modeled as an entity instead of just a relationship?

Answer: Appointment is considered an entity because it has its own attributes (Date, Time, Reason, AppointmentID) that would otherwise have no place to be stored.

2. What would be lost or break if Appointment were not an entity?

ANSWER: If Appointment were not an entity, those attributes would not exist on their own, which would cause trouble if a patient has multiple visits with the same doctor and could create time conflicts for appointments with the same doctor.